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Note

The number π plays a very important role in both physics and mathematics. It is an irrational number which cannot be expressed in finite decimal form. However, we can approximate π as ≈ 3.14 or $\frac{22}{7}$. But we should not forget that 3.14 or $\frac{22}{7}$ is only an approximation to π not equal to π.

2.11.5 Angular displacement

Consider a particle revolving around a point O in a circle of radius r (Figure 2.45). Let the position of the particle at time t = 0 be A and after time t, its position is B.

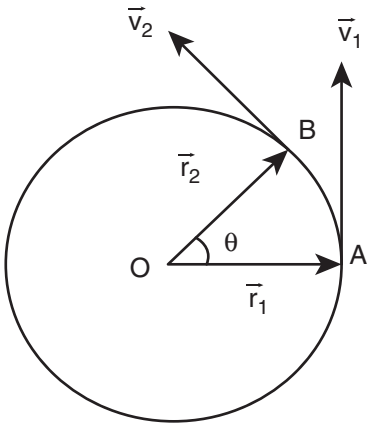


Figure 2.45 Angular displacement

Then,

The angle described by the particle about the axis of rotation (or center O) in a given time is called angular displacement.

i.e., angular displacement = $\angle AOB = \theta$

The unit of angular displacement is radian.

The angular displacement (θ) in radian is related to arc length S (AB) and radius r as

$$\theta = \frac{S}{r}, \quad \text{or} \quad S = r\theta$$

Angular velocity ($\vec{\omega}$)

The rate of change of angular displacement is called angular velocity.

If θ is the angular displacement in time t, then the angular velocity ω is

$$\omega = \lim_{\Delta t \to 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}$$

The unit of angular velocity is radian per second (rad s^{-1}). The direction of angular velocity is along the axis of rotation following the right hand rule. This is shown in Figure 2.46.

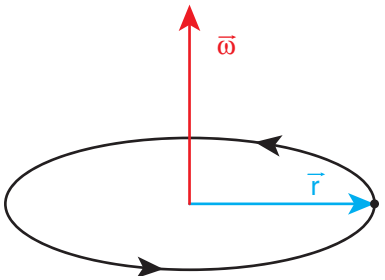


Figure 2.46 Direction of angular velocity

i) Angular acceleration (α)

The rate of change of angular velocity is called angular acceleration.

$$\vec{\alpha} = \frac{d\vec{\omega}}{dt}$$

The angular acceleration is also a vector quantity which need not be in the same direction as angular velocity.